Chapter 3 exercises

1. Prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ by the element method, and identify the use of extensionality.
2. In $U = \{1, \ldots, 30\}$, let A, B, C be multiples of two, three, and five. Find the union size and the number belonging to exactly one set.
3. Roll two independent fair six-sided dice. Let A mean sum seven and B mean first die four. Find $P(A), P(A|B)$ and determine independence.
4. Under the assumed testing model, prevalence is 2%, sensitivity 95%, and specificity 90%. Use Bayes' formula for disease probability given a positive test. Give expected counts for 10,000 people and explain how the prior enters the denominator.
5. A variable X has values 0,1,2 with probabilities $c, 2c, 3c$. Find $c, EX, \mathrm{Var}(X)$.
6. Can positive-probability events satisfy $P(A|B) > P(A)$ but $P(B|A) < P(B)$? Prove the relation between the directions of these comparisons with marginals.
7. Let $A = \{a, b\}, B = \{b, c\}$, with distinct symbols. List $\mathcal{P}(A) \cap \mathcal{P}(B)$ and $\mathcal{P}(A \cup B) \setminus (\mathcal{P}(A) \cup \mathcal{P}(B))$. Prove generally that $\mathcal{P}(A) \subseteq \mathcal{P}(B) \iff A \subseteq B$.
8. For fair mutually independent bits x_1, x_2, x_3, let $A = \{x_1 = x_2\}, B = \{x_2 = x_3\}, C = \{x_1 = x_3\}$. Prove pairwise independence and decide mutual independence.
9. Twelve distinguishable components contain four defects. Uniformly sample three without replacement. Find the complete PMF, mean, and variance of defect count X. Recompute variance using covariances of three position indicators.
10. On uniform $\Omega = \{00, 01, 10, 11\}$, let I_1, I_2 be the bits and I_3 their XOR. Find the PMF, mean, and variance of $S = I_1 + I_2 + I_3$, comparing with Binomial$(3, 1/2)$. Do equal means and variances prove equal distributions?
11. Suppose $P(Y = 0) = 2/3, P(Y = 1) = 1/3$, with $X|Y = 0$ Bernoulli$(1/4)$ and $X|Y = 1$ Bernoulli$(3/4)$. Find $P(X = 1)$, then both within-group and between-group terms of total variance, checking directly.
12. Let X be uniform on $\{-1, 0, 1\}$ and $Y = X^2$. Find covariance, determine independence, and compute $\mathrm{Var}(X + Y)$. Explain the logical difference between zero covariance and independence.
13. Let first-success trial T be Geometric$(1/3)$, supported from one. Find mean, variance, and $P(T \geq 9)$, comparing Markov and Chebyshev bounds for this one-sided tail. Prove memorylessness $P(T > m + n|T > m) = P(T > n), m, n \geq 0$.

Chapter 3 exercises (continued)

14. Joint masses at $(0,0), (0,1), (1,0), (1,1)$ are $1/8, 3/8, 3/8, 1/8$. Find marginals, $P(X=1|Y=1)$, covariance, and $\mathrm{Var}(X+Y)$. Check the sum's PMF.
15. On $\Omega = \{1, 2, \ldots\}$, let $P(\{n\}) = 1/[n(n+1)]$ and $X(n) = n$. Prove this is a probability distribution, find $P(X \geq m)$, and prove EX is not finite. Why is $E[X^2] - (EX)^2$ unavailable as a finite-variance calculation?
16. On $\Omega = \{a, b, c\}$, masses are $1/2, 1/2, 0$. Put $A = \{a\}, B = \{c\}$. Is B empty? Is $P(A|B)$ defined here? Are A, B independent? Apply definitions, distinguishing sets and probabilities rather than relying on everyday "impossibility."

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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https://github.com/xinhi6716/computer-science-self-study

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